

APPROVAL NOTE

Repair of Nozzle N1 on Vessel V-1201

Ref. No.: MRPL/I&R/APR/2023/0088

Date: 22 Nov 2023

Department: Inspection and Reliability

Category: Capital repair

Originated By: S. Rajagopal, Inspection Engineer

Priority: High

1. Background

During the external inspection round of October 2023, corrosion under insulation was confirmed at the N1 inlet nozzle of vessel V-1201. Stripping of the insulation revealed general wall loss over an area of approximately 180 mm by 220 mm on the nozzle neck, with a minimum remaining thickness of 8.6 mm against a required minimum of 9.4 mm.

2. Recommendation

It is recommended that the affected section of the N1 nozzle neck be replaced with a new forged section to the original specification during the CDU-I shutdown scheduled for February 2024, and that the insulation system be renewed with a moisture barrier to the current CUI standard.

3. Financial Implication

The estimated cost of the repair, comprising material, fabrication, non destructive examination and insulation renewal, is INR 42.6 lakh. Provision exists within the approved shutdown budget for CDU-I under head CAP/2024/CDU/11.

4. Approval

The recommendation at clause 2 above, and the expenditure of INR 42.6 lakh at clause 3, were approved by the General Manager, Technical Services, on 24 November 2023. Execution is entrusted to the Maintenance Department under the supervision of the Inspection Engineer.

Prepared By

Reviewed By

Approved By